

COMP364 Quiz 6

Question 1. Masked self-attention (5 points)

In the phrase *masked self-attention*, briefly describe the meaning of “masked” and “self”.

Question 2. Single-head attention operator (5 points)

List the major quantities employed by a single-head attention operator, in the approximate order of computation.

Question 3. Multi-head attention operator (5 points)

Describe the difference between single-head and a multi-head attention in one sentence.

Question 4. Attention layer (5 points)

List the major quantities employed by an attention layer, in the order of computation.

Question 5. Transformer (5 points)

List the major quantities and components of a transformer neural network architecture, in their order of computation.

Question 6. Training targets (5 points)

A certain transformer uses lower case letters from the English alphabet as its only tokens. The following sequence is its only training data:

yymfndifhdjsjshfgkgdjdvbkfrbsbivb

The transformer employs a block size of 5, and it is being trained using batches of size 2. One particular batch of training data employs the inputs

$$x = \begin{pmatrix} \text{hdjsj} \\ \text{jdvbk} \end{pmatrix}.$$

For this batch, what value of the targets y should be employed?

Question 7. Predictions and loss calculation

A transformer uses the vocabulary $\{\mathbf{a}, \mathbf{b}\}$ with IDs $\{\mathbf{a} : 0, \mathbf{b} : 1\}$. It processes a sequence of length 3, producing the final logits

$$\begin{pmatrix} 3.1 & -0.5 \\ 0.2 & 0.7 \\ -1.1 & -0.1 \end{pmatrix}.$$

- (a) **(5 points)** What are the predicted probabilities for the next token, and which token will be output?
- (b) **(5 points)** Suppose instead that the same logits above were produced by training on a single input sequence $x = \mathbf{aba}$ and target $y = \mathbf{abb}$. What is the contribution of the final token to the cross-entropy loss?

Total number of points: **40**